

Example 3.15.

Let $\mathcal{B}$ be the blackbox for the rational function $\mathcal{B}(x, y) = \frac{x+5y}{3+xy} \in \mathbb{Z}_p(x, y)$ in $lex(x > y)$ that we wish to recover using Kaltofen-Yang algorithm.

Since we have two variables, we will evaluate the blackbox $\mathcal{B}$ at $[2^i, 3^i]$, where $0 \leq i \leq n$. Let number of points $t = 5$. The first step is to find the univariate image of $\mathcal{B}(x, y)$ in $\mathbb{Z}_p(x)$

$$\begin{aligned} \phi : \mathbb{Z}_p(x, y) &\longrightarrow \mathbb{Z}_p(x) \\ x &\longmapsto x \\ y &\longmapsto \beta(x - \sigma_1) + \sigma_2 \end{aligned}$$

$$T(x) = \mathcal{B}(x, \beta(x - \sigma_1) + \sigma_2) = \frac{ff(x, \beta(x - \sigma_1) + \sigma_2)}{gg(x, \beta(x - \sigma_1) + \sigma_2)}$$

$$\text{Let } \sigma^i = \begin{bmatrix} \sigma_1^i \\ \sigma_2^i \end{bmatrix} = \begin{bmatrix} 2^i \\ 3^i \end{bmatrix} \quad \text{and } \beta = 7$$

$$\implies T(x) = \mathcal{B}(x, 7(x - \sigma_1^i) + \sigma_2^i) = \frac{ff(x, 7(x - \sigma_1^i) + \sigma_2^i)}{gg(x, 7(x - \sigma_1^i) + \sigma_2^i)} \quad (1)$$

For $i = 0$ we'll have $\sigma^0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Substituting σ^0 in equation 1.

$$T(x) = \mathcal{B}(x, 7(x - 1) + 1) = \frac{ff(x, 7(x - 1) + 1)}{gg(x, 7(x - 1) + 1)}$$

Let $\alpha = [\alpha_1 \ \alpha_2 \ \alpha_3 \ \alpha_4 \ \alpha_5]^T = [1 \ 2 \ 3 \ 4 \ 5]^T$ Now, we want to find out $T(\alpha_i)$ where $1 \leq i \leq 5$ by evaluating the blackbox $\mathcal{B}$ at $[\alpha_i, 7(\alpha_i - 1) + 1]$.

$$T(\alpha_i) = \mathcal{B}(\alpha_i, 7(\alpha_i - 1) + 1) = \frac{ff(\alpha_i, 7(\alpha_i - 1) + 1)}{gg(\alpha_i, 7(\alpha_i - 1) + 1)}$$

Now we can use Maximal quotient rational function reconstruction algorithm to recover the uni-

Table 1: Computed values of σ_1 , σ_2 and $T(\alpha)$

α	$(\alpha, \beta(\alpha - \sigma_1(i)) + \sigma_2)$	$T(\alpha) = \mathcal{B}(\alpha, 7(\alpha - 1) + 1)$
1	[1, 1]	55
2	[2, 8]	36
3	[3, 15]	15
4	[4, 22]	33
5	[5, 29]	95

variate image of the numerator and denominator separately. The modulus polynomial is

$$\overline{m} = (x - 1)(x - 2)(x - 3)(x - 4)(x - 5) = x^5 + 92x^4 + 85x^3 + 96x^2 + 60x + 94.$$

The interpolating polynomial is

$$u(x) = 52x^4 + 4x^3 + 66x^2 + 45x + 102,$$

Table 2: MQRFR computations for input polynomials $\overline{m}(x)$ and $u(x)$

i	q_i	r_i	t_i
0	—	$x^5 + 92x^4 + 85x^3 + 96x^2 + 60x + 94$	0
1	$35x + 32$	$52x^4 + 4x^3 + 66x^2 + 45x + 102$	1
2	$52x + 21$	$x^3 + 47x^2 + 79x + 40$	$72x + 75$
3	$21x^2 + 95x + 44$	$51x + 11$	$x^2 + 45x + 31$
4	—	91	$86x^4 + 30x^3 + 59x^2 + 69x + 102$

Applying the maximal quotient rational function reconstruction algorithm (MQRFR) to $\overline{m}(x)$ and $u(x)$ modulo 107 yields the following sequence of quotients q_i , remainders r_i , and auxiliary polynomials t_i :

At the step where $\deg(q_i)$ is maximal ($i = 3$), we identify

$$\mathcal{B}(x) = 51x + 11, \quad g(x) = x^2 + 45x + 31.$$

These are the univariate images of the rational multivariate function we are trying to recover.

$$T(x) = \frac{51x + 11}{x^2 + 45x + 31}$$

For this example, we will reuse the alpha values, so our modulus polynomial is the same for each iteration, only the interpolating polynomial u changes. For $i = 1$ we'll have $\sigma^1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. Substituting σ^1 in equation 1.

$$T(x) = \mathcal{B}(x, 7(x - 2) + 3) = \frac{ff(x, 7(x - 2) + 3)}{gg(x, 7(x - 2) + 3)}$$

For $i = 2$ we'll have $\sigma^2 = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$. Substituting σ^2 in equation 1.

$$T(x) = \mathcal{B}(x, 7(x - 4) + 9) = \frac{ff(x, 7(x - 4) + 9)}{gg(x, 7(x - 4) + 9)}$$

For $i = 3$ we'll have $\sigma^3 = \begin{bmatrix} 8 \\ 27 \end{bmatrix}$. Substituting σ^3 in equation 1.

$$T(x) = \mathcal{B}(x, 7(x - 8) + 27) = \frac{ff(x, 7(x - 8) + 27)}{gg(x, 7(x - 8) + 27)}$$